InternNova Week 6
Complete Project Guide

Retail Sales Data Analytics & Business Intelligence | Atanu Manna

What is already completed

The uploaded UCI Online Retail Excel file has been read and analyzed. A cleaned transaction CSV, monthly/country/product/customer summaries, descriptive statistics, correlation matrix, audit JSON, and six charts have been generated. The report uses real values from the supplied dataset.

Verified item	Result
Original dataset rows	541,909
Exact duplicate rows removed	5,268
Positive-sales rows analyzed	524,878
Positive-sales revenue	£10,642,110.80
Unique invoices	19,960
Customers with ID	4,338
Unique product codes	3,922

Project files

Download the complete ZIP package separately. It includes the data, Python code, generated charts, report files, Power BI guide, Git instructions and presentation. The native Power BI .pbix and published GitHub repository still need to be created.

Step 1 — Extract the project ZIP

1. Download Week6_Retail_Sales_Analytics_COMPLETE_WORKING.zip from the ChatGPT link.
2. Right-click the ZIP in Windows File Explorer → Extract All.
3. Open the extracted Week6_Retail_Sales_Analytics folder.

Step 2 — Check project contents

Confirm that data/raw/Online Retail.xlsx exists, and that data/processed/ contains cleaned_sales.csv, monthly_sales.csv, country_sales.csv, product_sales.csv, customer_sales.csv, descriptive_statistics.csv, correlation_matrix.csv and analysis_audit.json. The visuals folder should contain six PNG charts.

Step 3 — Run Python again (optional)

Install Python 3.10 or newer if required. Open the project folder in File Explorer, type cmd in the address bar, then press Enter. Run these commands:

```
python -m pip install -r requirements.txt
```

```
python src/analyze_retail.py
```

If `python` is not recognized, try `py -m pip install -r requirements.txt` and `py src/analyze_retail.py`. If an error appears, save a screenshot and fix the error before continuing.

Step 4 — Build the real Power BI dashboard

1. Open Power BI Desktop.
2. Home → Get data → Text/CSV.
3. Select data/processed/cleaned_sales.csv and choose Load.
4. Verify InvoiceDate is Date/Time; Quantity is Whole Number; UnitPrice and LineRevenue are Decimal Number; invoice/product/customer IDs are Text.

Create the following measures using Modeling → New measure. Replace `cleaned_sales` if the table has a different name:

```
Total Revenue = SUM(cleaned_sales[LineRevenue])
```

```
Total Invoices = DISTINCTCOUNT(cleaned_sales[InvoiceNo])
```

```
Unique Customers = DISTINCTCOUNT(cleaned_sales[CustomerID])
```

```
Units Sold = SUM(cleaned_sales[Quantity])
```

```
Average Invoice Revenue = DIVIDE([Total Revenue], [Total Invoices], 0)
```

Step 5 — Add visuals and slicers

Create three KPI cards: Total Revenue, Total Invoices, Unique Customers. Add at least four visuals: (1) monthly revenue line chart using YearMonth and LineRevenue; (2) top products bar chart using Description and LineRevenue; (3) revenue by Country bar chart; (4) revenue by weekday column chart. Add slicers for Country, InvoiceDate and Description or StockCode. Use the title Online Retail Sales Performance (2010–2011) and format revenue as GBP (£).

Save the actual report as powerbi/Retail_Sales_Dashboard.pbix. Capture screenshots from Power BI Desktop showing the full dashboard and at least one slicer interaction. The included PNG charts are Python charts, not Power BI screenshots.

Step 6 — Create and push GitHub repository

On GitHub, create a public repository named retail-sales-analytics. If you already have a local README, create the remote repository without adding another README. Open Command Prompt inside the project folder and run:

```
git init
git branch -M main
git add README.md requirements.txt .gitignore
git commit -m "chore: add project structure and documentation"

git remote add origin https://github.com/YOUR-USERNAME/retail-sales-analytics.git
git push -u origin main
```

Replace YOUR-USERNAME with your real GitHub username. Then make two more meaningful commits after adding the pipeline and outputs:

```
git add src
git commit -m "feat: add retail data cleaning and EDA pipeline"
git push

git add data/processed visuals powerbi reports presentation README.md
git commit -m "docs: add analysis outputs and dashboard documentation"
git push
```

Check commits using `git log --oneline`. Do not claim a commit or push unless it succeeded. Do not put passwords or tokens in public files.

Step 7 — Finish report and presentation

Use the included report and slide drafts. Include the six generated charts, actual verified results, a screenshot from the real Power BI dashboard, five or more evidence-based insights, at least two recommendations, dataset citation, limitations and your GitHub URL. The presentation should explain the problem, dataset, cleaning, EDA, dashboard, insights and recommendations.

Verified findings from the uploaded dataset

• Highest-revenue country in the analyzed positive-sales records: United Kingdom (£9,001,744.09).

• Top product by line revenue: DOTCOM POSTAGE (£206,248.77).

• Highest-revenue month: 2011-11 (£1,503,866.78).

• Lowest-revenue month: 2011-02 (£522,545.56); partial months may affect comparison.

• Highest revenue weekday: Thursday (£2,199,292.57); lowest: Sunday (£806,790.78).

The source data are historical (2010–2011). Revenue is not profit because the dataset does not include costs. Treat these as historical analysis, not current market results.

Final submission checklist

■ Analysis outputs and six charts verified
■ Actual .pbix dashboard saved
■ Genuine Power BI screenshots inserted
■ At least three meaningful Git commits pushed
■ README, report and presentation completed
■ Repository URL opens successfully
■ Submit one permitted format: GitHub link, Google Drive link, or PDF upload.

Status note

This guide and analysis are prepared. A native .pbix dashboard and publishing/pushing to your GitHub account require actions in Power BI Desktop and GitHub on your computer. Complete those before submitting this as a fully finished project.